Complex Differentiability, Cauchy–Riemann Equations, and Holomorphic Rigidity

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§1.1 One difference quotient must survive every planar direction

Let $U \subseteq \mathbb{C}$ be open and let $f : U \to \mathbb{C}$. The complex derivative at $z_0 \in U$ is

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h},$$

provided the limit exists. The increment h approaches zero through the whole plane. This is much stronger than asking for two one-sided real limits.

Write

$$z = x + iy, \qquad f(z) = u(x, y) + iv(x, y).$$

If f is complex differentiable at $z_0 = x_0 + iy_0$, then the quotient has the same limit along the real and imaginary axes.

Along $h = t \in \mathbb{R}$,

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

Along $h = it$,

$$\begin{aligned} f'(z_0) &= \lim_{t \to 0} \frac{u(x_0, y_0 + t) - u(x_0, y_0) + i\big(v(x_0, y_0 + t) - v(x_0, y_0)\big)}{it} \\ &= v_y(x_0, y_0) - iu_y(x_0, y_0). \end{aligned}$$

Equality gives the Cauchy–Riemann equations

$$\boxed{u_x = v_y, \qquad u_y = -v_x.}$$

The derivative can then be written in either form:

$$f' = u_x + iv_x = v_y - iu_y.$$

§1.2 Cauchy–Riemann is sufficient under real differentiability

The equations alone at one point do not automatically imply complex differentiability. A clean sufficient hypothesis is real differentiability.

Proposition 1.2.1

Suppose $u, v : \mathbb{R}^2 \to \mathbb{R}$ are differentiable at (x_0, y_0) and satisfy the Cauchy–Riemann equations there. Then $f = u + iv$ is complex differentiable at $z_0 = x_0 + iy_0$.

Proof. Let $h = \Delta x + i\Delta y$. Real differentiability gives

$$\Delta u = u_x \Delta x + u_y \Delta y + r_1(h),$$
$$\Delta v = v_x \Delta x + v_y \Delta y + r_2(h),$$

where

$$\frac{|r_1(h)| + |r_2(h)|}{|h|} \to 0.$$

Using $u_x = v_y$ and $u_y = -v_x$,

$$\begin{aligned} \Delta f &= \Delta u + i\Delta v \\ &= (u_x + iv_x)(\Delta x + i\Delta y) + r_1(h) + ir_2(h). \end{aligned}$$

Hence

$$\frac{\Delta f}{h} = u_x + iv_x + \frac{r_1(h) + ir_2(h)}{h} \longrightarrow u_x + iv_x.$$

□

If the first partial derivatives are continuous in a neighborhood, real differentiability is automatic. Thus the familiar C^1 version is:

$$f \text{ holomorphic} \iff u_x = v_y, \quad u_y = -v_x.$$

§1.3 Wirtinger derivatives isolate the holomorphic part

Define

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y).$$

Since

$$x = \frac{z + \bar{z}}{2}, \qquad y = \frac{z - \bar{z}}{2i},$$

these operators treat z and $\bar{z}$ as formally independent coordinates:

$$\partial_z z = 1, \quad \partial_z \bar{z} = 0, \quad \partial_{\bar{z}} z = 0, \quad \partial_{\bar{z}} \bar{z} = 1.$$

For $f = u + iv$,

$$2\partial_{\bar{z}} f = (u_x - v_y) + i(v_x + u_y).$$

Therefore the Cauchy–Riemann equations are exactly

$$\boxed{\partial_{\bar{z}} f = 0.}$$

Under real differentiability, this is equivalent to holomorphicity.

The operator $\partial_{\bar{z}}$ measures failure of holomorphicity. For example,

$$\partial_{\bar{z}} \bar{z} = 1,$$

so $\bar{z}$ is nowhere holomorphic. Also

$$|z|^2 = z\bar{z}, \qquad \partial_{\bar{z}} |z|^2 = z.$$

Hence $|z|^2$ is complex differentiable only at $z = 0$, and it is not holomorphic on any neighborhood of that point.

§1.4 The Jacobian must be a rotation followed by a scaling

The real derivative of $f = u + iv$ is the Jacobian matrix

$$Df = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}.$$

Under the Cauchy–Riemann equations,

$$Df = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad a = u_x, \quad b = v_x.$$

This is precisely the real matrix of multiplication by

$$f'(z_0) = a + ib.$$

If $f'(z_0) \neq 0$, write

$$f'(z_0) = \rho e^{i\theta}.$$

Then

$$Df(z_0) = \rho \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Infinitesimally, f scales every direction by ρ and rotates every direction by θ. Thus a holomorphic map with nonzero derivative preserves oriented angles. This is the local meaning of conformality.

At a critical point $f'(z_0) = 0$, the first-order map collapses. For instance, $f(z) = z^m$ near 0 multiplies angles by m but has zero linear part when $m \geq 2$.

§1.5 Cauchy's integral formula turns boundary values into derivatives

The rigidity of holomorphic functions does not follow from the Cauchy–Riemann equations alone by a short algebraic argument. The decisive analytic input is Cauchy's integral formula.

If f is holomorphic on a neighborhood of the closed disk

$$\overline{D(a, R)},$$

then for $|z - a| < R$,

$$\boxed{f(z) = \frac{1}{2\pi i} \int_{|\zeta - a| = R} \frac{f(\zeta)}{\zeta - z}\, d\zeta.}$$

Differentiating under the integral sign gives

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{|\zeta - a| = R} \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta.$$

One complex derivative therefore forces derivatives of every order.

The same formula yields the Cauchy estimate. If

$$M_R = \max_{|\zeta - a| = R} |f(\zeta)|,$$

then

$$|f^{(n)}(a)| \leq \frac{n! M_R}{R^n}.$$

Boundary control becomes quantitative control of all derivatives in the interior.

§1.6 Holomorphic functions are analytic

For $|z-a|<R$, expand the kernel as a geometric series:

$$\begin{aligned}\frac{1}{\zeta - z} &= \frac{1}{\zeta - a}\frac{1}{1-\frac{z-a}{\zeta-a}} \\ &= \sum_{n=0}^{\infty} \frac{(z-a)^n}{(\zeta-a)^{n+1}}.\end{aligned}$$

The series converges uniformly on the integration circle for $|z-a|<R$. Substitution into Cauchy's formula gives

$$f(z) = \sum_{n=0}^{\infty} c_n (z-a)^n,$$

where

$$c_n = \frac{1}{2\pi i}\int_{|\zeta-a|=R} \frac{f(\zeta)}{(\zeta-a)^{n+1}}\,d\zeta = \frac{f^{(n)}(a)}{n!}.$$

Thus

$$\boxed{f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(z-a)^n.}$$

Holomorphicity implies analyticity. This is the sharp contrast with smooth real functions such as

$$g(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

whose Taylor series at 0 vanishes identically although the function does not.

§1.7 Zeros are isolated unless the whole function vanishes

Suppose f is holomorphic near a and not identically zero. Its Taylor series has a first nonzero coefficient:

$$f(z) = c_m(z-a)^m + c_{m+1}(z-a)^{m+1} + \cdots, \qquad c_m \neq 0.$$

Factor

$$f(z) = (z-a)^m g(z), \qquad g(a) = c_m \neq 0.$$

By continuity, g is nonzero near a, so a is an isolated zero of multiplicity m.

This proves the identity theorem. If two holomorphic functions agree on a set with a limit point inside a connected domain, their difference has non-isolated zeros and must vanish identically.

The chain of rigidity is now explicit:

$$\begin{aligned}\partial_{\bar z} f = 0 &\longrightarrow \text{Cauchy formula} \longrightarrow \text{power series}, \\ &\longrightarrow \text{isolated zeros and unique continuation}.\end{aligned}$$

The operator $\partial_{\bar z}$ is the local differential equation; Cauchy's formula supplies the global analytic mechanism that makes its solutions rigid.